Photorealistic Enhancement with Paired Data

Morandi Francesca
283200@studenti.unimore.it
Artificial Intelligence Engineering
Unimore

Melotti Virginia
283715@studenti.unimore.it
Artificial Intelligence Engineering
Unimore

Fiorani Riccardo
343485@studenti.unimore.it
Artificial Intelligence Engineering
Unimore

Abstract

Photorealistic enhancement is an image-to-image translation task, that consists in making rendered or synthetic images more realistic.

The challenge of enhancing image realism is crucial for various applications, such as gaming and virtual and augmented reality, where more realistic visuals increase user immersion and engagement, as well as scientific visualization, where accurate and lifelike representations of complex data can aid in analysis.

In our study we propose an approach that aims to solve the synthetic-to-real problem by creating a paired dataset of synthetic and real-world images. The paired dataset is then used to finetune a pre-trained Pix2Pix Generative Adversarial Network and test it. We then compare its performance with a Pix2Pix trained on a randomly paired data and with a GAN model trained on unpaired data.

1. Introduction

The generation of photorealistic images has been the main focus of computer graphics for half a century and as the demand for highly realistic visuals has increased, the need for advanced techniques to narrow the gap between synthetic and real-world images has become more pressing.

In the last few years, with the advent of deep learning, the field of image-to-image translation, in particular synthetic to real translation, has seen big improvements.

Traditionally, the challenge to bridge the gap between synthetic and real data has been addressed through the use of Generative Adversarial Networks (GANs) [2]. Generative Adversarial Networks (GANs) are a type of deep learning model composed of two neural networks: a generator and a discriminator. The generator creates synthetic data, such as images, that mimic real data, while the discriminator evaluates whether the data is real or generated. The two networks are trained together in a process known as adversarial training, where the generator tries to fool the discriminator, and the discriminator works to accurately distinguish between real and synthetic data. Over time, this competition drives the generator to produce increasingly realistic outputs, making GANs powerful tools for tasks like image generation and enhancement.

GANs used for synthetic-to-real image translation, such as CycleGAN [9] or SimGAN [8], are typically designed to work with unpaired data. This means that the model learns to translate images from one domain (synthetic) to another (real) without requiring a direct one-to-one correspondence between the images in the two domains. Although unpaired data is easy to acquire, it has some limitations. Without paired data, the model may struggle to capture fine details and subtle variations that are crucial for achieving true photorealism. The absence of direct guidance can lead to inconsistencies in the generated images, such as inaccurate lighting, texture, or perspective.

In this study, we propose an alternative approach that exploits paired data to train a GAN, specifically the Pix2Pix model. Our idea is that by using a paired dataset, where each synthetic image has a corresponding real-world counterpart, the model may learn more precise mappings between the synthetic and real domains. This could lead to more accurate and realistic translations, as the model would have access to specific examples of how synthetic features should appear in the real world. Since we didn't have a perfect correspondence between the synthetic and real images, we decided to pair the images ourselves. After some pre-processing, we fed the images into a feature extraction algorithm, and given the extracted features, we computed the most similar pairs and created our paired dataset.

Our procedure acts like a middle ground between the unpaired and paired approach. We'll compare the performance of the Pix2Pix with our paired data to its unpaired counterpart with a CycleGAN, as well as to a Pix2Pix trained on randomly paired data. Our goal is to see if to solve the problem of phorealistic enhancement of synthetic data, the best course of action is the traditional unpaired approach or our paired approach.

2. Related Work

Many previous studies have investigated the use of GANs to enhance synthetic data for machine learning applications.

Zhu et al. (2017) introduced the CycleGAN framework, which enables unsupervised image-to-image translation thanks to a cycle-consistency loss, which ensures that an image translated from the source domain to the target domain can be converted back to the original source domain without losing information.

Shrivastava et al. (2017) developed SimGAN, which learns to map simulated images to real ones, thereby generating more realistic images that improve the performance of downstream tasks like object detection.

2.1. State of the art models

The current state of the art models for image synthesis and photorealism enhancement are the StyleGAN models (2022) [1]. Unlike traditional GANs that typically generate images from a latent vector in a straightforward manner, StyleGAN employs a style-based generator architecture. This approach uses multiple levels of abstraction to control different aspects of the generated images, such as texture and high-level features. In particular, the best model is StyleGAN3, that introduces improvements in image fidelity and reduces common artifacts seen in earlier models, achieving excellent FID scores.

Another model that excels in generating high-quality images and has shown competitive FID scores in various benchmarks is Stable Diffusion [7]. This Diffusion model operates in a latent space, by first mapping the image to a lower-dimensional latent space and then performing the diffusion process there.

2.2. GANs designed for paired data

Pix2Pix (2018) [4] is one of the first GANs designed for paired image-to-image translation tasks. Pix2Pix translates images from one domain to another by learning the mapping between input-output pairs.

3. Data

3.1. GTA5

The GTA5 dataset contains 24966 synthetic images with pixel level semantic annotation. The images have been rendered using the open-world video game Grand Theft Auto 5 and are all from the car perspective in the streets of American-style virtual cities. There are 19 semantic classes which are compatible with the ones of Cityscapes dataset.

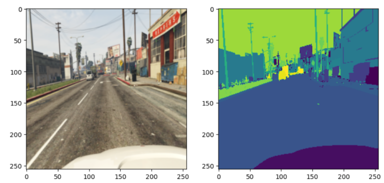

Figure 1. GTA5 sample

3.2. Cityscapes

The Cityscapes Dataset contains a diverse set of stereo video sequences recorded in street scenes from 50 different cities, with high quality pixel-level annotations of 5000 frames in addition to a larger set of 20000 weakly annotated frames.

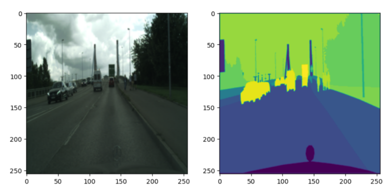

Figure 2. Cityscapes sample

3.3. Our Dataset

The paired dataset used in this study is composed of 2500 synthetic images from the GTA5 dataset and 2500 corresponding real-world images from the Cityscapes dataset.

To create the paired dataset:

- Firstly, the images from the two datasets were normalized in the pixel range [0,1] and resized in the right resolution for the feature extraction algorithm (224x224)
- Histogram Matching was applied to the GTA images to align their color distribution with the Cityscapes images.
- A pre-trained Dinov2-small model was employed to extract the features from both datasets. Cosine similarity was calculated to find the closest matches, resulting in paired GTA-to-Cityscapes images.
- The final dataset was split into training (0,8) and testing (0,2) sets, maintaining the correspondence of similar pairs.
- Additional preprocessing steps to the paired dataset included Bilateral Filtering to reduce noise in the GTA images and random Perspective Transformations for data augmentation in the training set.

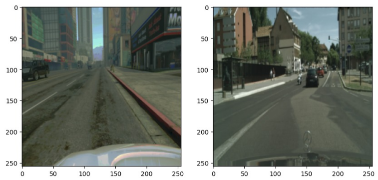

Figure 3. GTA-to-Cityscapes sample, where synthetic GTA image is on the left and corresponding Cityscapes image is on the right

4. Methods

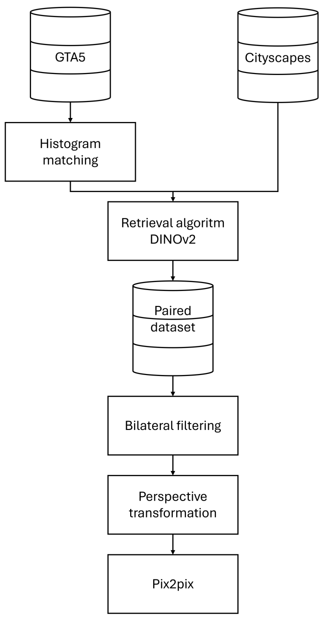

Figure 4. Project pipeline

4.1. Image-processing operators

Histogram matching : The first thing we did after resizing and normalizing the images of both datasets, was to apply Histogram matching on the GTA5 images to adjust the color distribution of synthetic images to match the histogram of the real images. This helps in better feature extraction and pairing during the Retrieval process, since the overall intensity distributions of the images will be more consistent with each other.

We didn't implement it from scratch, but used the function `match_histogram` of the module `skimage`, that accepts the np.arrays of the synthetic and reference real image.

Bilateral filtering After the Retrieval algorithm, as a preprocessing step on the paired dataset, we applied Bilateral filtering on the GTA images to reduce artificial noise, while preserving important details such as edges. This way, the synthetic images can better match the natural quality of the real Cityscapes images, which remain unaltered to ensure that they provide a true-to-life reference for the model.

We used the `cv2.bilateralFilter` function, considering a 5x5 pixel neighborhood, a sigmaColor (how much intensity differences are allowed to affect the filtering) of 25 and a sigmaSpace (spatial extent of the filter's effect) of 25.

4.2. Geometric based algorithm

After creating, preprocessing and splitting the paired dataset, we applied on half the training set some random Perspective transformations. The images (GTA and Cityscapes) belonging to the same sample undergo the same transformation to maintain the correspondence, while the transformations among different samples have different distortion angles and ranges. This process creates dataset augmentation, increasing the dataset variability, improving the model robustness to various viewing angles and positions and enhancing model generalization.

4.3. Retrieval algorithm

To collect the best pairs for our paired dataset:

- we first used a pretrained DINOv2-small model [6] to extract the features from the GTA5 and Cityscapes images. DINOv2-small is a light and efficient variant of the DINOv2 model, that uses the Vision Transformer (ViT) architecture. The model is pre-trained on large and diverse datasets, such as COCO and Imagenet, which helps it capture general visual features. DINOv2-small needs the images to be in the 224X224 resolution.

  We chose to extract the features from the final transformer layer of the model. This layer contains high-level, abstract representations that are ideal for comparing images in a retrieval task.

- the similarity between the extracted feature vectors was then measured with the Cosine similarity [5]. We selected for each GTA image, the Cityscapes image that gave the highest Cosine similarity.

Figure 5. Sample from the training set, with bilateral filtering and perspective transformation applied to it

4.4. Deep-learning based component

Our final paired dataset was then used to finetune and test a Pix2Pix model [3]. Pix2Pix was chosen first of all, because it excels in scenarios where the datasets are paired, allowing the model to learn a direct mapping between input and output. In second place, it is a suitable choice for many image enhancement scenarios because of its versatility. In fact, it can be used for a variety of image-to-image translation tasks, such as turning sketches into photos or converting day images to night images.

Pix2Pix is based on a conditional GAN architecture, made of a U-Net generator and a PatchGAN discriminator. The term "conditional" is due to the fact that it generates output images based on a given input image, while unconditional GANs generate samples from a distribution without conditioning on any specific input.

- The U-Net generator, with its encoder-decoder structure and skip connections, is responsible for producing the output image given an input image. It learns to map the input image to the desired output image, by retaining both high-level features and fine details.
- The PatchGAN discriminator's role is to classify whether patches of the image are real or fake. It guides the generator to produce more realistic images.
- During training the model combines the adversarial loss between the generator and discriminator with the L1 reconstruction loss, ensuring that the generated images not only appear realistic but also accurately reflect the input images.

$$\mathcal{L}_{\text{Pix2Pix}}(G, D) = \mathcal{L}_{\text{GAN}}(G, D) + \lambda \cdot \mathcal{L}_{L1}(G) \quad (1)$$

Due to the limited dimension of our dataset, we chose to finetune a Pix2Pix model pretrained on the Day2Night dataset. The Day2Night dataset is composed of urban images taken from Cityscapes, where each daytime image has a corresponding nighttime image. Although our paired GTA-to-Cityscapes dataset is different from Day2Night, the two datasets share common visual elements like urban landscapes and varying lighting conditions. The general features learned from Day2Night might help in generating high-quality images in the new domain, and moreover, the variety of textures, colors, and lighting conditions contained in Day2Night could be beneficial for generalization. To train and test the model, we had to resize the images to the right resolution (256x256) and denormalize them back to the original pixel values.

4.5. Pix2Pix Training

We trained the Pix2Pix model under two different scenarios, both involving finetuning a pretrained model on the Day2Night dataset. In each case, we used the pretrained generator, randomly initialized the discriminator, and trained for 100 epochs on the new dataset.

1. **Case 1:** Finetuning on the paired GTA-to-Cityscapes dataset with the following settings:
   - **Learning Rate:** 0.0001 for the first 25 epochs, 0.0002 for the next 25 epochs, and a decreasing rate from 0.0001 for the remaining 50 epochs.
   - **Optimizer:** Adam ($\beta_1 = 0.5$).
   - **Network Architecture:** U-Net generator with 256 filters (`unet_256`) and basic discriminator with 3 layers (`basic`).
   - **Normalization:** Batch normalization.
   - **Dropout:** Enabled.
   - **Loss Function:** Vanilla GAN loss with L1 loss for the generator ($\lambda_{L1} = 100.0$).
2. **Case 2:** Finetuning on a 'randomly paired dataset'. We used the same parameters as Case 1 and randomly generated a paired dataset without the application of any retrieval algorithm.

4.6. CycleGAN Training

For comparison between the paired and traditional unpaired approaches, we trained a CycleGAN model with the following parameters:

- **Learning Rate:** 0.0002.
- **Optimizer:** Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$).
- **Network Architecture:** ResNet generator with 9 blocks (`resnet_9blocks`) and basic discriminator (`basic`).
- **Normalization:** Instance normalization.
- **Loss Function:** Least Squares GAN loss with cycle consistency loss ($\lambda_A = 10.0$, $\lambda_B = 10.0$) and identity loss ($\lambda_{identity} = 0.5$).
- **Training Strategy:** 100 epochs with linear decay over 100 epochs.
- **Pooling:** Image buffer size of 50.
- **Initialization:** Normal distribution with gain 0.02.
- **Dropout:** Disabled.

4.7. Training with and without Mixed Precision

4.7.1 Mixed Precision

In modern deep learning, optimizing computational efficiency and resource utilization is critical, especially when working with large models and high-resolution datasets. Mixed precision training has emerged as an effective technique to address these challenges. By combining 16-bit (half-precision) and 32-bit (single-precision) floating-point operations, mixed precision training allows models to benefit from reduced memory usage and faster computation times while maintaining the accuracy and stability of traditional training methods.

In this study, we applied mixed precision training to both Pix2Pix and CycleGAN models to evaluate its impact on performance, particularly in terms of GPU memory usage and execution time. Leveraging PyTorch's `torch.cuda.amp` module, we automatically managed the precision of operations within our models.

In this setup:

- **Precision Handling**: The `torch.cuda.amp` module was used to automatically manage the casting of operations to either 16-bit or 32-bit precision. This allowed most of the model's operations to run in half-precision while maintaining full precision where necessary to avoid loss of accuracy.

- **Gradient Scaler**: We utilized `amp.GradScaler()` to scale the loss during backpropagation, which helps prevent underflow issues that can occur with 16-bit precision. The scaler dynamically adjusts the scaling factor during training to optimize the stability of the gradients.

- **Batch Size**: With reduced memory usage from mixed precision, we were able to increase the batch size from 16 to 32, improving the efficiency of the training process without running into memory limits on the GPU.

- **Learning Rate Adjustments**: To accommodate the changes in gradient accumulation behavior introduced by mixed precision, the learning rate was scaled by a factor of 1.5. This adjustment was determined based on preliminary experiments to ensure stable convergence.

4.7.2 Pix2Pix and CycleGAN Training Results

To evaluate the impact of mixed precision training on performance and resource usage, we compared the results of both Pix2Pix and CycleGAN models under mixed precision and full precision configurations:

Pix2Pix Training:

- **Mixed Precision Training:**
  - **Batch Size:** 32
  - **GPU Memory Usage:** 5 GB
  - **Execution Time per Epoch:** 2 minutes
- **Full Precision Training:**
  - **Batch Size:** 32
  - **GPU Memory Usage:** 6 GB
  - **Execution Time per Epoch:** 2 minutes 30 seconds

CycleGAN Training:

- **Mixed Precision Training:**
  - **Batch Size:** 4
  - **GPU Memory Usage:** 8 GB
  - **Execution Time per Epoch:** 10 minutes
- **Full Precision Training:**
  - **Batch Size:** 4
  - **GPU Memory Usage:** 10 GB
  - **Execution Time per Epoch:** 11 minutes

The results showed that the mixed precision approach significantly increased the training efficiency and reduced the GPU memory usage and execution time across both models, without compromising the quality of the generated images. Because of the limited computational resources we had at our disposal, we chose to carry out the mixed precision training in the case of the CycleGAN, whose training time with the full precision resulted too long. This enabled us to train larger batch sizes and reduce the overall training time.

4.8. Inference

4.8.1 Metrics

The metrics we used during testing to compare the results are:

- FID: (Fréchet Inception Distance) measures the similarity between two sets of images, typically real images and generated images. It does this by comparing the statistics (mean and covariance) of the feature representations of the images, extracted by a pre-trained neural network.

  A low FID score indicates that the generated images are close to the real images in terms of their distribution and have good quality and variation.

For calculating the FID in our project, we utilized the `pytorch-fid` module.

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}\left(\Sigma_r + \Sigma_g - 2\sqrt{\Sigma_r\Sigma_g}\right)$$

- SSIM: (Structural Similarity Index) measures the similarity between two images, focusing on perceived changes in structural information while being less sensitive to absolute differences like brightness or contrast. SSIM evaluates three aspects: luminance, contrast, and structure, comparing these between the reference and the generated image. The SSIM score ranges from -1 to 1, with 1 indicating perfect structural similarity.

  A higher SSIM score means that the generated image is structurally closer to the reference image and the important features and textures have been preserved.

  The module we used to compute the SSIM is `skimage.metrics`.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

- PSNR: (Peak Signal-to-Noise Ratio) measures the peak error between the reference image and the generated image, providing an estimate of the image's fidelity and quantifying how much detail or quality was lost in the generation. PSNR calculation is based on the mean squared error (MSE) between the two images. It gives a logarithmic scale score (in decibels, dB) where a higher PSNR indicates better quality of the generated image.

  A high PSNR value (from 30 to 40 dB) indicates that the generated image has low distortion and is close to the original image.

  In our project, we chose to implement the PSNR formula from scratch.

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}}\right)$$

4.8.2 Results

In the following table are shown the results of the phase of testing and evaluation:

Model	Dataset	FID	SSIM	PSNR
Pix2Pix	paired	169.51	0.346	28.21
Pix2Pix	random	170.52	0.340	28.10
CycleGAN	unpaired	163.68	0.634	28.12

(a) synthetic image (b) generated image (c) real image

Figure 6. Testing sample generated by the Pix2Pix trained on the GTA-to-Cityscapes paired dataset

(a) synthetic image (b) generated image (c) real image

Figure 7. Testing sample generated by the Pix2Pix trained on the randomly paired dataset

(a) synthetic image (b) generated image

Figure 8. Testing sample generated by the CycleGAN trained on the unpaired GTA and Cityscapes datasets

4.8.3 Results analysis

- By observing the results obtained by the Pix2Pix trained with our GTA-to-Cityscapes paired dataset, we can notice that with a FID of 169.51, a SSIM of 0.346 and a PSNR of 28.21, the model fails to achieve a satisfactory level of quality and realism. While the generated images roughly resemble street scenes from a visual standpoint, they lack significant detail, exhibit some noise and degree of distortion and are significantly different from the real images from a distributional and structural standpoint. Moreover, the generated images differ considerably in terms of their configuration and fundamental elements from the reference synthetic images, which should have been made more realistic without being completely altered.

The reason of this outcome can be attributed to the limited nature of our dataset. In fact, due to scarce computational resources, we had to create our paired dataset by selecting the optimal pairs from just a few thousand of images from both GTA5 and Cityscapes datasets. This lead to a dataset formed by not perfectly correspondent synthetic-to-real pairs, that probably 'confused' the Pix2Pix model during training.

- Comparing the results obtained by the Pix2Pix on our paired data with the Pix2Pix trained on a randomly paired dataset, we can notice a slight improvement in all the metrics.

  This means that although the results reached with our dataset are not excellent, the improvement over random pairing indicates that the retrieval model successfully captured some degree of relationship between input and output images, and the model benefits from having more accurate input-output pairs during training.

- The comparison of the results between the Pix2Pix model trained on our paired dataset and the CycleGAN model trained on the unpaired dataset reveals that the latter produces superior images in terms of quality for all the evaluated metrics. Notably, the SSIM score of 0.634 indicates that images generated by CycleGAN exhibit greater structural similarity to real images. Furthermore, CycleGAN, with its unsupervised and unpaired approach, is able to enhance the realism of the synthetic images while preserving greater fidelity to their original composition and fundamental elements compared to Pix2Pix.

5. Future Work

Because of the limited nature of our paired dataset, we weren't able to reach satisfactory results in training the Pix2Pix model to achieve true photorealism in the generated images. Consequently, in our future works we aim to expand the dataset and create more correspondent synthetic-to-real pairs.

Moreover, we want to modify the Pix2Pix architecture to exploit the segmentation of the images to better understand the spatial relations between the synthetic and real images.

6. Conclusion

In this study, we aimed to determine whether the traditional unpaired data approach or our paired method would yield better results in achieving photorealistic enhancement of synthetic data taken from the GTA5 dataset.

The results demonstrated that the Pix2Pix model trained on our paired GTA5-to-Cityscapes dataset struggled to capture the fine details and subtleties necessary for true photorealism. Additionally, when comparing the results to the CycleGAN model trained on unpaired data, the Pix2Pix model did not outperform its unpaired counterpart.

The limitations observed in our paired approach suggest that, while pairing synthetic and real images may offer a more structured approach to GAN training for image-to-image translation, a non perfectly correspondent paired dataset is not sufficient for achieving high-quality photorealistic results.

References

[1] Amit H. Bermano, Rinon Gal, Yuval Alaluf, Ron Mokady, Yotam Nitzan, Omer Tov, Or Patashnik, and Daniel Cohen-Or. State-of-the-art in the architecture, methods and applications of stylegan, 2022. 2

[2] Antonia Creswell, Tom White, Vincent Dumoulin, Kai Arulkumaran, Biswa Sengupta, and Anil A. Bharath. Generative adversarial networks: An overview. *IEEE Signal Processing Magazine*, 35(1):53–65, 2018. 1

[3] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. pix2pix. https://github.com/junyanz/pix2pix, 2017. Accessed: 2024-09-02. 4

[4] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. Image-to-image translation with conditional adversarial networks, 2018. 2

[5] Ylber Januzaj and Artan Luma. Cosine similarity – a computing approach to match similarity between higher education programs and job market demands based on maximum number of common words. *International Journal of Emerging Technologies in Learning (iJET)*, 17:258–268, 06 2022. 3

[6] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. Dinov2: Learning robust visual features without supervision, 2024. 3

[7] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022. 2

[8] Simiao Yu, Hao Dong, Felix Liang, Yuanhan Mo, Chao Wu, and Yike Guo. Simgan: Photo-realistic semantic image manipulation using generative adversarial networks. In *2019 IEEE International Conference on Image Processing (ICIP)*, pages 734–738. IEEE, 2019. 1

[9] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros. Unpaired image-to-image translation using cycle-consistent

adversarial networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2017. 1